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Nullstellensatz **1**

Nullstellensatz

$$\{\text{prime ideals of } K[X] \text{ with finite residue}\} = \{\text{max ideals of } K[X]\}$$

$\Longrightarrow$: $I \subset K[X]$ is prime $\Longrightarrow$ $K[X]/I$ is integral.

Let $K[X]/I = K(\alpha_1, ..., \alpha_n)$ then by the exercise $K[X]/I$ is a field $\Longrightarrow$ I is maximal.

$\Longleftarrow$: $M \subset K[X]$ be maximal $\Longrightarrow$ $K[X]/M = K(\alpha_1, ..., \alpha_n)$ by Nullz.

In commutative rings, max ideals are also prime $\Longrightarrow$ max ideals of $K[X]$ are finite residues of K and prime.